

Department of Computer Science and Engineering

Introduction to Natural Language Processing

Study Guide

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1. Introduction to Natural Language Processing (NLP)

1.1 Introduction

Natural Language Processing (NLP) is a branch of Artificial Intelligence (AI) that focuses on enabling computers to understand, interpret, process, and generate human language in a meaningful way. NLP combines concepts from computer science, linguistics, machine learning, and deep learning.

Human languages such as English, Hindi, Gujarati, French, and others are highly complex due to ambiguity, contextual meanings, grammar variations, and semantic structures. NLP aims to bridge the communication gap between humans and machines.

Examples of NLP systems include:

- Chatbots
- Machine Translation Systems
- Voice Assistants
- Sentiment Analysis Systems
- Text Summarization Tools
- Search Engines

1.2 Goals of NLP

The primary goals of NLP are:

1. To enable computers to understand natural language.
2. To facilitate human-computer interaction.
3. To extract meaningful information from text.
4. To generate human-like language.
5. To automate language-based tasks.

1.3 Components of NLP

NLP mainly consists of two components:

1. Natural Language Understanding (NLU)

NLU focuses on enabling machines to understand the meaning and intent of human language.

2. Natural Language Generation (NLG)

NLG deals with generating meaningful and grammatically correct text.

1.4 NLP Workflow

Diagram: Basic NLP Workflow

Input Text → Preprocessing → Feature Extraction → NLP Model → Output

Example:

Sentence: "Students are learning NLP."

- Tokenization → [Students, are, learning, NLP]
- POS Tagging → Noun, Verb, Verb, Noun
- Semantic Analysis → Understanding meaning

2. Evolution of NLP

NLP has evolved through three major waves:

6. First Wave – Rationalism
7. Second Wave – Empiricism
8. Third Wave – Deep Learning

2.1 First Wave: Rationalism (1950s–1980s)

The first wave of NLP focused on symbolic and rule-based approaches.

Characteristics

- Handcrafted grammar rules
- Knowledge-based systems
- Logical reasoning
- Linguistic frameworks

Techniques

- Rule-Based Systems
- Expert Systems
- Syntax Trees
- Semantic Networks

Advantages

- Human understandable
- Strong linguistic foundation

Limitations

- Difficult to scale
- Cannot handle ambiguity effectively
- Requires extensive manual effort

Example

Rule:

IF sentence contains “is” THEN classify as declarative sentence.

2.2 Second Wave: Empiricism (1990s–2010)

This wave introduced statistical and machine learning methods.

Characteristics

- Corpus-based learning
- Statistical analysis
- Probabilistic models
- Large text datasets

Techniques

- Hidden Markov Models (HMM)
- Naïve Bayes
- N-grams
- Statistical Machine Translation

Applications

- POS Tagging
- Named Entity Recognition
- Sentiment Analysis

Advantages

- Better scalability
- Data-driven learning

Limitations

- Sparse data problem
- Limited semantic understanding

2.3 Third Wave: Deep Learning (2010–Present)

The third wave introduced deep neural networks for language understanding.

Characteristics

- Deep neural architectures
- Automatic feature learning
- Context-aware processing
- Large-scale language models

Important Architectures

- Recurrent Neural Networks (RNN)
- Long Short-Term Memory (LSTM)
- Convolutional Neural Networks (CNN)
- Transformers

Modern NLP Models

- BERT
- GPT
- T5
- LLaMA

Advantages

- High accuracy
- Contextual understanding
- Better semantic representation

Limitations

- Large computational requirements
- Data dependency
- Ethical concerns

2.4 Comparison of NLP Waves

Feature	First Wave	Second Wave	Third Wave
Approach	Rule-Based	Statistical	Deep Learning
Data Requirement	Low	Medium	Very High
Scalability	Low	Medium	High
Context Understanding	Poor	Moderate	Excellent
Accuracy	Moderate	Good	Very High

3. Difficulties and Importance of NLP

3.1 Difficulties in NLP

Natural language is highly complex and ambiguous.

1. Ambiguity

Words may have multiple meanings.

Example:

“Bank” may refer to:

- Financial institution
- River bank

2. Context Dependency

Meaning depends on surrounding words.

Example:

“I saw her duck.”

3. Grammar Variations

Different writing styles and sentence structures create complexity.

4. Sarcasm and Emotion

Machines struggle to detect emotions and sarcasm.

Example:

“Great! Another assignment.”

5. Data Sparsity

Limited training data affects model performance.

6. Real-world Noise

Social media text often contains:

- Misspellings
- Slang
- Abbreviations
- Emojis

3.2 Importance of NLP

NLP has become essential in modern computing.

Major Importance Areas

9. Language Understanding
10. Machine Translation
11. Information Extraction
12. Text Summarization
13. Question Answering
14. Dialogue Systems
15. Speech Recognition
16. Knowledge Representation

3.3 Real-Life Applications

- Google Translate
- Siri
- Alexa
- ChatGPT
- Email Spam Filtering
- Recommendation Systems

4. Basic NLP Operations

NLP processing involves several stages.

4.1 NLP Pipeline

Diagram: NLP Processing Pipeline

Input Text → Lexical Analysis → Parsing → Semantic Analysis → Discourse Integration → Pragmatic Analysis

4.2 Word Level Analysis (Lexical Analysis)

This stage focuses on analyzing individual words.

Operations

- Tokenization
- Stemming

- Lemmatization
- Stop-word Removal
- POS Tagging

Example

Sentence:

“Students are learning NLP.”

Tokens:

[Students, are, learning, NLP]

4.3 Syntactic Analysis (Parsing)

Parsing identifies grammatical structure.

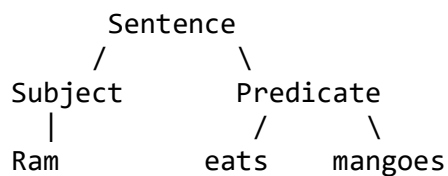
Example

Sentence:

“Ram eats mangoes.”

Subject → Verb → Object

Parse Tree



4.4 Semantic Analysis

Semantic analysis determines meaning.

Tasks

- Word Sense Disambiguation
- Semantic Role Labeling
- Semantic Parsing

Example

“The bank approved the loan.”

Meaning of “bank” = Financial institution

4.5 Discourse Integration

Discourse processing studies relationships among sentences.

Example

“Rahul bought a laptop. He loves it.”

“He” refers to Rahul.

4.6 Pragmatic Analysis

Pragmatics studies intended meaning.

Example

“Can you open the window?”

Literal meaning: Asking ability

Actual meaning: Request to open window

5. Word Level Analysis (Lexical Analysis)

Lexical analysis is the first stage in NLP.

5.1 Tokenization

Breaking text into smaller units called tokens.

Example

Sentence:

“Natural Language Processing is interesting.”

Tokens:

[Natural, Language, Processing, is, interesting]

5.2 Stemming

Stemming reduces words to root forms.

Examples

- Playing → Play
- Studies → Studi
- Connected → Connect

Popular Stemmer

Porter Stemmer

5.3 Lemmatization

Lemmatization converts words into meaningful base forms.

Examples

- Running → Run
- Better → Good

Difference Between Stemming and Lemmatization

Feature	Stemming	Lemmatization
Accuracy	Low	High
Output	Root Form	Dictionary Form
Linguistic Knowledge	Not Required	Required

5.4 Stop-word Removal

Commonly occurring words are removed.

Examples:

- is
- the
- a
- an
- and

6. Syntactic Analysis (Parsing)

Syntactic analysis determines grammatical structure.

6.1 Types of Parsing

1. Top-Down Parsing

Starts from root node.

2. Bottom-Up Parsing

Starts from input words.

6.2 Dependency Parsing

Shows relationships between words.

Example

Sentence:

“Students study NLP.”

study → root

Students → subject

NLP → object

6.3 Constituency Parsing

Divides sentence into phrases.

Example

[NP Students] [VP study NLP]

6.4 Applications of Parsing

- Grammar checking
- Machine translation
- Question answering
- Speech recognition

7. Semantic Analysis

Semantic analysis extracts meaning from text.

7.1 Objectives

- Understand word meaning
- Identify semantic relationships
- Interpret sentence meaning

7.2 Semantic Relationships

Synonymy

Words with similar meaning.

Example:

Big ↔ Large

Antonymy

Opposite meaning.

Example:

Hot ↔ Cold

Hyponymy

Specific relationship.

Example:

Rose → Flower

7.3 Semantic Role Labeling

Identifies semantic roles in sentence.

Example

“Rahul opened the door.”

- Rahul → Agent
- Door → Object

8. Word Sense Disambiguation (WSD)

Word Sense Disambiguation identifies correct word meaning.

8.1 Need for WSD

Many words have multiple meanings.

Example

“Bat”

- Flying mammal
- Cricket equipment

8.2 Approaches to WSD

1. Knowledge-Based Approach

Uses dictionaries and ontologies.

2. Supervised Learning

Uses labeled training data.

3. Unsupervised Learning

Uses clustering methods.

8.3 Applications of WSD

- Machine Translation
- Search Engines
- Question Answering
- Text Mining

9. Discourse Processing

Discourse processing studies sentence-level relationships.

9.1 Importance

It improves text coherence and contextual understanding.

9.2 Anaphora Resolution

Identifies references.

Example

“Priya bought a car. She likes it.”

- She → Priya
- It → Car

9.3 Coherence in Text

Text should logically flow between sentences.

Example

Incorrect:

“Ramesh studied hard. The weather was sunny.”

Correct:

“Ramesh studied hard. Therefore, he scored good marks.”

10. Part of Speech (PoS) Tagging

POS tagging assigns grammatical labels to words.

10.1 Common POS Tags

Tag	Meaning	Example
NN	Noun	dog
NNS	Plural Noun	dogs
VB	Verb	eat
JJ	Adjective	beautiful
RB	Adverb	quickly
PRP	Pronoun	he

10.2 Example of POS Tagging

Sentence:

“Ram eats mangoes.”

Word	Tag
Ram	NNP
eats	VBZ
mangoes	NNS

10.3 POS Tagging Approaches

Rule-Based Approach

Uses grammar rules.

Statistical Approach

Uses probabilistic models.

Machine Learning Approach

Uses supervised learning.

Deep Learning Approach

Uses RNNs and LSTMs.

10.4 Applications of POS Tagging

- Machine Translation
- Speech Recognition
- Grammar Checking
- Information Extraction

11. Natural Language Inference and Information Retrieval

11.1 Natural Language Inference (NLI)

NLI determines logical relationships between sentences.

Relationships

- Entailment
- Contradiction

- Neutral

Example

Sentence 1:

“A man is playing cricket.”

Sentence 2:

“A person is playing a sport.”

Relationship → Entailment

11.2 Information Retrieval (IR)

Information Retrieval focuses on retrieving relevant information.

Applications

- Search Engines
- Digital Libraries
- Recommendation Systems

11.3 Components of IR System

Diagram: Information Retrieval System

Documents → Indexing → Query Processing → Retrieval Model → Ranked Results

11.4 Retrieval Models

Vector Space Model (VSM)

Represents documents as vectors.

BM25

Ranking algorithm used in search engines.

Language Models

Probabilistic retrieval methods.

11.5 Evaluation Metrics

Precision

Measures relevant retrieved documents.

Recall

Measures total relevant documents retrieved.

F1 Score

Harmonic mean of precision and recall.

Formula:

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

12. Applications of NLP

NLP has applications in many domains.

12.1 Sentiment Analysis

Analyzes emotions in text.

Example:

“This product is excellent.” → Positive sentiment

12.2 Machine Translation

Automatically translates text.

Example:

English → Hindi

“Good Morning” → “□□□□□□□□”

12.3 Chatbots and Virtual Assistants

Examples:

- Siri
- Alexa

- Google Assistant
- ChatGPT

12.4 Text Summarization

Generates concise summaries.

Applications:

- News summarization
- Research paper summarization

12.5 Question Answering Systems

Answers user queries automatically.

Applications:

- Educational systems
- Customer support

12.6 Healthcare Applications

- Medical report analysis
- Disease prediction
- Clinical text mining

12.7 Financial Applications

- Fraud detection
- Market sentiment analysis
- Risk analysis

12.8 Social Media Analytics

Analyzes public opinions and trends.

Applications:

- Political analysis
- Brand monitoring
- Public feedback

13. Summary

Natural Language Processing is an important field of Artificial Intelligence that enables machines to process and understand human language. NLP has evolved from rule-based systems to modern deep learning architectures such as Transformers and Large Language Models.

The unit covered:

- Evolution of NLP
- NLP Challenges
- NLP Processing Pipeline
- Lexical, Syntactic and Semantic Analysis
- Word Sense Disambiguation
- Discourse Processing
- POS Tagging
- Information Retrieval
- Applications of NLP

Modern NLP systems are widely used in healthcare, education, social media, search engines, and intelligent virtual assistants.